

XR Cricothyroidotomy and Intraosseous Access: A Preliminary Study on Usability and Presence between VR/MR and Hand Tracking/Controller Interactions

Dale Button*
Primary Care Paramedic
and Advanced Paramedic
Durham College

Alvaro Uribe-Quevedo†
Faculty of Business and IT
Ontario Tech University.

Adam Dubrowski‡
Faculty of Health Sciences
Ontario Tech University



Figure 1: Virtual (VR) and mixed reality (MR) Tibial intraosseous access (IO) and Cricothyroidotomy (Crico) with hand tracking (HT) and VR controller (VRC) inputs.

ABSTRACT

Consumer-level extended reality (XR) devices and available software development tools have sparked content creation in training and education. Current research highlights the benefits of using XR for learning, and the gaps pertaining to motor skills, since consumer-level devices lack proper task and instrument representation that is essential for developing transferable skills. Our paper, presents a preliminary study on usability, cognitive load, and presence using an off-the-shelf XR device to better understand the user experience effects when combining virtual and mixed reality with hand tracking and controller inputs that will inform future work for designing effective properly represented interactions.

Index Terms: Extended reality, hand tracking, task trainers.

1 INTRODUCTION

Immersive technologies have evolved from devices capable of providing virtual, augmented, or mixed reality experiences (VR, AR, and MR, respectively) to extended reality (XR) devices capable of providing VR, AR, and MR on one device [2]. This recent trend is providing exciting immersive training and education opportunities that allow learners and trainees to learn and develop skills in highly engaging simulated scenarios, while leveraging real-world elements. Research centered on XR has led to a better understanding of the effects associated with safe exposure to virtual realistic experiences, health risks caused by overexertion, repetitive movements, disorientation, discomfort, and motion sickness that these devices pose for training and education [7].

This paper presents a preliminary study examining usability, cognitive load, and presence to better understand user experience when practicing virtual and mixed reality (VR and XR, respectively) cricothyroidotomy (Crico) and intraosseous tibial access

(IO), using a VR controller (VRC) and hand-tracking (HT). Crico and IO procedures are commonly practiced interventions that require extensive training with high-end simulators that allow trainees to practice the procedure in controlled environments [9]. However, the use of advanced simulators and their associated costs pertaining to acquisition, maintenance, personnel training, curriculum adaptation, specialized infrastructure, and availability to ensure adequate access to trainees limits access in cases where scheduling, maintenance, or pandemics impede their use [6]. Such limitations have sparked the design of cost-effective solutions using XR [1] and 3D printing [4].

2 MATERIALS AND METHODS

A within-subjects study approved under Ontario Tech University's Research Ethics Board #15128 included the following conditions: i) VR with VRC, ii) VR with HT, iii) MR with VRC, and iv) MR with HT was conducted.

2.1 Study Procedures

Participants completed the following for each condition: i) read the instructions for the procedure, ii) performed the task, and iii) completed the System Usability Scale (SUS) [3] for usability, the Raw NASA Task Load Index (RTLX) [5] for cognitive load, and the Presence Questionnaire (PQ) [10] for presence, followed by an open question to share thoughts about the experience.

2.2 Apparatus

The Crico and IOTA scenarios were developed using Unity's game engine version 2022.3.16f1, Meta's XR interaction software development kit (SDK), Meta's haptic SDK, and existing 3D models to populate the scene. In order to rapidly prototype the scenarios, examples from Meta XR's SDK and building blocks were repurposed to create four scenarios using VR and MR, both with HT and VRC support (Fig. 1).

3 PRELIMINARY RESULTS

Five participants, four women, and one man with a Health Sciences background, volunteered for our preliminary study. Only one participant reported having previous experience with VR, Crico and

*e-mail: dale.button@durhamcollege.ca

†e-mail: alvaro.quevedo@ontariotechu.ca

‡e-mail: adam.dubrowski@ontariotechu.ca

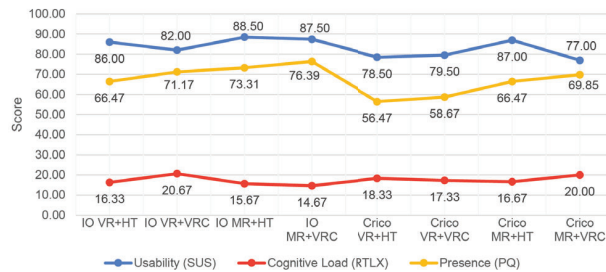


Figure 2: Usability, cognitive load, and presence results.

IO by developing 3D-printed simulators. Although another participant indicated that they were familiar with VR, they did not use it frequently. Fig. 2 presents the results of usability, cognitive load, and presence according to condition. Usability scores were greater than 68, which is the minimum threshold for marginally acceptable usability. Similarly, scores greater than 80 are considered excellent. The highest usability was associated with the IO MR with HT ($M = 88.50$, $SD = 12.94$), followed by the IO MR with VRC ($M = 87.50$, $SD = 12.24$), the Crico MR with HT ($M = 87.00$, $SD = 13.85$), the IO VR with HT ($M = 86.00$, $SD = 11.80$), and the IO VR with VRC ($M = 82$, $SD = 15.55$). Next, conditions with good usability (SUS scores between 70 and 80) include the Crico VR with VRC ($M = 79.50$, $SD = 18.40$), the Crico VR with HT ($M = 78.50$, $SD = 18.50$), and crico MR with VRC ($M = 77$, $SD = 20.18$).

The cognitive load between 0 and 9 is classified as low, 10 - 29 as medium, 30 - 49 as somewhat high, 50 - 79 as high, and 80 - 100 as very high [8]. The RTLX scores indicate that all conditions had a medium cognitive load, with the lowest score obtained by the IO MR with VRC ($M = 14.67$, $SD = 5.19$), followed by the IO MR with HT ($M = 15.67$, $SD = 6.41$), the IO VR with HT ($M = 16.33$, $SD = 7.58$), the Crico MR with HT ($M = 16.67$, $SD = 5.13$), the Crico VR with VRC ($M = 17.33$, $SD = 7.87$), the Crico VR with HT ($M = 18.33$, $SD = 5.13$), the Crico MR with VRC ($M = 20.00$, $SD = 9.64$), and the IO VR with VRC ($M = 20.67$, $SD = 11.46$).

The IO MR with VRC had the highest presence score ($M = 76.39$, $SD =$), followed by the IO MR with HT ($M = 73.31$, $SD =$), the IO VR with VRC ($M = 71.17$, $SD = 6.82$), the Crico MR with VRC ($M = 69.85$, $SD = 9.79$), the IO VR with HT ($M = 66.47$, $SD = 11.92$) was tied to Crico MR with HT ($M = 66.47$, $SD = 12.06$), and finally, the Crico VR with VRC ($M = 58.67$, $SD = 10.6$) and the Crico VR with HT ($M = 56.47$, $SD = 8.80$) had the lowest scores.

3.1 Open Feedback

The participants provided positive feedback, in addition to highlighting the clarity of the interactions and appreciating the possibility of getting closer to where drilling and scalpel slicing occur. All participants enjoyed HT, as it felt natural and easy to perform. However, they preferred the VRC for drilling, as it felt like holding and activating a drill. Despite positive feedback, participants struggle with fine movement tasks, such as aiming the drill, operating the scalpel, and placing pins on anatomical landmarks. Additional feedback received from participants included suggestions for including visual elements representative of a hospital, in addition to increasing visual realism for the IO scenario.

4 DISCUSSION

This paper presents a preliminary study exploring the differences in usability, cognitive load, and presence between VR and MR with standard hand tracking and controller interactions available in the Meta XR SDK when practicing cricothyroidotomy and intraosseous tibial access procedures. Although exploratory in nature and limited by the small number of participants, we believe that our work

provides a foundation for better understanding how VR and MR in conjunction with input devices, including VR controllers and hand tracking, affect usability, cognitive load, and presence, unlike studies that focus only on educational value with standard VR controller interactions. Furthermore, the preliminary results indicate that conditions with higher usability had lower cognitive load and vice versa, and that the combination of MR with HT and VRC provided greater usability for the IO task. Regarding presence, the use of the VRC for IO was higher than HT for Crico. We believe that the VR controller's shape provided greater realism, while the manipulation of the virtual scalpel was more challenging, as the controller did not properly represent it, thus creating a disparity between what is seen and what is felt. The difficulties related to the lack of tactile feedback with HT could have influenced the tasks, and further analysis is needed to identify how tracking limitations impact performance.

Regardless of VR and MR, the participants were very active when performing the procedures, sitting and standing up to examine the virtual patients and performing the tasks. Furthermore, participants expressed a clear preference for MR because they did not feel isolated as in VR, a feature highlighted to be relevant when collaborating with others. Additionally, they also expressed their preference for using the VRC for the IO task, and HT for Crico.

Future work will focus on adding instructional and game design elements to evaluate the development of cognitive and motor skills toward the evaluation of transferrable skills, in addition to a larger user study.

ACKNOWLEDGMENTS

This work was supported by NSERC RGPIN-2018-05917.

REFERENCES

- [1] S. F. Alves, M. Sivanathan, J. Micallef, B. Gino, M. Mnaymneh, A. Dubrowski, and A. Uribe-Quevedo. Cricothyroidotomy simulator: A makerspace and augmented reality approach. In *2022 IEEE Games, Entertainment, Media Conference (GEM)*, pp. 1–4. IEEE, 2022. 1
- [2] S. Barteit, L. Lanfermann, T. Bärnighausen, F. Neuhaus, C. Beiersmann, et al. Augmented, mixed, and virtual reality-based head-mounted devices for medical education: systematic review. *JMIR Serious Games*, 9(3):e29080, 2021. 1
- [3] J. Brooke. Sus: a retrospective. *Journal of Usability Studies*, 8(2):29–40, 2013. 1
- [4] B. Gino, R. Ingino, A. Raza, A. Dubrowski, and A. Uribe-Quevedo. Authoring cricothyroidotomy teaching with augmented and virtual reality. In *2023 IEEE 3rd Colombian BioCAS Workshop*, pp. 1–6. IEEE, 2023. 1
- [5] S. G. Hart. Nasa-task load index (nasa-tlx); 20 years later. In *Proceedings of the Human Factors and Ergonomics Society Annual meeting*, vol. 50, pp. 904–908. Sage publications Sage CA: Los Angeles, CA, 2006. 1
- [6] D. S. Hippe, R. A. Umoren, A. McGee, S. L. Bucher, and B. W. Bresnahan. A targeted systematic review of cost analyses for implementation of simulation-based education in healthcare. *SAGE Open Medicine*, 8:2050312120913451, 2020. 1
- [7] A. D. Kaplan, J. Cruick, M. Endsley, S. M. Beers, B. D. Sawyer, and P. A. Hancock. The effects of virtual reality, augmented reality, and mixed reality as training enhancement methods: A meta-analysis. *Human Factors*, 63(4):706–726, 2021. 1
- [8] D. Restuputri, S. Fatimah, A. Mubin, et al. Air traffic control work system design to improve operator performance with workload approach and safety concept. *Jurnal Sistem dan Manajemen Industri*, 6(2):200–214, 2022. 2
- [9] V. Scali and J. Espinosa. Critical skills mannequin training for cricothyrotomy. *Mathews Journal of Emergency Medicine*, 5(2):1–5, 2020. 1
- [10] B. G. Witmer and M. J. Singer. Measuring presence in virtual environments: A presence questionnaire. *Presence*, 7(3):225–240, 1998. 1